Chapter 31

Astronomy and cosmology

LEARNING INTENTIONS

In this chapter you will learn how to:

- understand the term luminosity as the total power of radiation emitted by a star
- recall and use the inverse square law for radiant flux intensity F in terms of the luminosity L of the source: $F = \frac{L}{4\pi d^2}$
- understand that an object of known luminosity is called a standard candle
- understand the use of standard candles to determine distances to galaxies
- recall and use Wien's displacement law $\lambda_{max} \propto \frac{1}{T}$ to estimate the peak surface temperature of a star
- use the Stefan-Boltzmann law $L = 4\pi\sigma r^2 T^4$
- use Wien's displacement law and the Stefan-Boltzmann law to estimate the radius of a star
- understand that the lines in the emission spectra from distant objects show an increase in wavelength from their known values
- use $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for the redshift of electromagnetic radiation from a source moving relative to an observer
- explain why redshift leads to the idea that the Universe is expanding
- recall and use Hubble's Law $v \approx H_0 d$ and explain how this leads to the Big Bang theory.

BEFORE YOU START

- Your knowledge of electromagnetic waves, including spectra, would be valuable in the understanding of this chapter.
- Can you recall intensity of a wave and its units?

- The idea of the Doppler effect of sound will be extended to spectra from distant stars. When is the observed wavelength shorter, or longer?

LOOKING INTO THE PAST

Figure 31.1 shows galaxies as seen through a powerful telescope. Each galaxy may have as many as 10^{11} stars, and there may be as many as 10^{11} galaxies in the Universe. Light from these galaxies has a finite speed: 3.0×10^8 m s^{-1} in a vacuum. These galaxies are so distant that light from them may have taken billions of years to reach us. So, what we have in this photograph is an image of the past. Andromeda is our closest galaxy. The light from this galaxy would take 2.3 million years to reach us. When we see this galaxy though a telescope, we are looking at its image from 2.3 million years ago! Just to put this into perspective, someone looking at the Earth from this galaxy now, would see a time when our ape-like ancestors roamed the planet.

In this chapter, we will deduce that galaxies further away from us are moving faster. This, in turns, implies that our Universe had a beginning–it was created some 14 billion years ago in an event known as the **Big Bang**. Since then, the fabric of the Universe has been stretching, carrying with it the galaxies.

Can you estimate the size and the mass of the Universe?

Figure 31.1: A cluster of distant galaxies; some created only a few million years after the creation of the Universe.

31.1 Standard candles

All the stars we see in the night sky are from our own galaxy–the Milky Way. Figure 31.2 shows stars in the constellation of Gemini. The stars do not look the same; they differ in brightness and colour. These stars are not all the same distance from us, and they do not all emit the same power. So, we cannot deduce their distance from just how bright they appear in the night sky.

Figure 31.2: Stars have different colours and brightness. Can you tell which star is the closest?

In astronomy, **luminosity** of a star is defined as the **total** radiant energy emitted per unit time. This is the same as the total power emitted by a star. In SI units, luminosity L is measured in W or J s^{-1}. The Sun is the nearest star to us, and astronomers have determined its luminosity to a high degree of accuracy. The luminosity of the Sun (solar luminosity), often written as $L_{\odot}$, is about 3.83×10^{26} W.

As you will see later, you can determine solar luminosity from the intensity of solar radiation reaching the Earth.

In astronomy, a **standard candle** is an astronomical object of known luminosity. Astronomers can determine the distance of a standard candle by measuring the intensity of the electromagnetic radiation arriving at the Earth.

Standard candles have been successfully used to determine the distance of far-flung galaxies. It is amazing that we can do this just by observing the starlight reaching us on Earth.

The two well-known standard candles are Cepheid variable stars and Type 1A supernovae.

Cepheid variable stars

In 1908, Henrietta Leavitt discovered that the brightness of Cepheid variable stars varied periodically, and the period of this variation was related to the average luminosity of the star. By measuring the period, astronomers could determine the luminosity of the star. The star’s distance could then be calculated from the observed radiant intensity at the Earth. Finding a Cepheid variable star in a distant galaxy meant that the distance of the galaxy itself could be calculated.

Type 1A supernovae

Type 1A supernovae stars implode rapidly towards the end of their lives, and scatter matter and energy out into space. This implosion event can be brighter than the galaxy itself. The luminosity of the star at the time of the implosion is always the same. From this, astronomers can estimate the star’s distance from the Earth.

31.2 Luminosity and radiant flux intensity

The Sun is the nearest star to the Earth. The second nearest star is Proxima Centauri, 4.0×10^{16} m away. Distances as large as 4.0×10^{16} m are extremely difficult to visualise–they are way beyond any of our day-to-day points of reference. So, astronomers tend to use an alternative unit for distance – the **light-year (ly)**. A light-year is the distance travelled by light in a vacuum in a time of one year. Therefore:

1 ly = speed of light in vacuum × one year in seconds

$$1 \text{ ly} \approx 3.00 \times 10^8 \times 365 \times 24 \times 3600$$
$$1 \text{ ly} \approx 9.5 \times 10^{15} \text{m}$$

Proxima Centauri is 4.2 ly away. It would take light from Proxima Centauri 4.2 years to reach us.

(Note: You do not need to know about light-years, but they help to visualise vast distances.)

Table 31.1 summarises some data on the brightest stars–do not forget that the Sun is a star too.

Rank order	Name of star	Distance / light-years	Temperature / K	Luminosity / $L_{\odot}$
1	Sun	1.58×10^{-5}	5800	1.0
2	Sirius	8.6	9900	25
3	Canopus	310	7000	1100
4	Alpha Centauri	4.4	5800	1.5
5	Arcturus	37	4300	170
6	Vega	25	9600	40

Table 31.1: Data on the six brightest stars, including the Sun. The luminosity is given in terms of the solar luminosity $L_{\odot}$; 1 $L_{\odot}$ = 3.83×10^{26} W.

We can see from Table 31.1 that the observed brightness of a star is linked to both its distance from the Earth and its luminosity. We would expect a luminous star, such as Canopus, to be bright in the night sky. Alpha Centauri is brighter in the night sky than Arcturus, not because of its luminosity, but because of its closeness to us. You will see later that the luminosity of a star depends not only on its surface temperature but also on its physical size.

Can we relate the brightness of a star to its luminosity? Yes, as long as we understand the underlying assumptions that:

- the power from the star is uniformly radiated through space
- there is negligible absorption of this radiated power between the star and the Earth.

With these assumptions, we can determine the intensity of electromagnetic radiation observed at the Earth.

The observed intensity is known as **radiant flux intensity** F. This is defined as the radiant power passing **normally** through a surface per unit area.

Figure 31.3 shows how F can be calculated for a star at a distance d from its **centre**.

$$\text{radiant flux intensity} = \frac{\text{power of star}}{\text{surface area of sphere}}$$

The power of the star is its luminosity L, and the surface area of a sphere is $4\pi d^2$.

Therefore:

$$F = \frac{L}{4\pi d^2}$$

The SI units for radiant flux intensity are W m^{-2}.

For a given star, the luminosity L is constant, so according to the equation, the radiant flux intensity F obeys an inverse square law with distance d. So, doubling the distance from the **centre** of the star ($2d$) will decrease F by a factor of 4, and tripling the distance ($3d$) will decrease F by a factor of 9, and so on.

You can demonstrate this inverse square law using a bright filament lamp and a light-meter in a darkened laboratory – see Practical Activity 31.1.

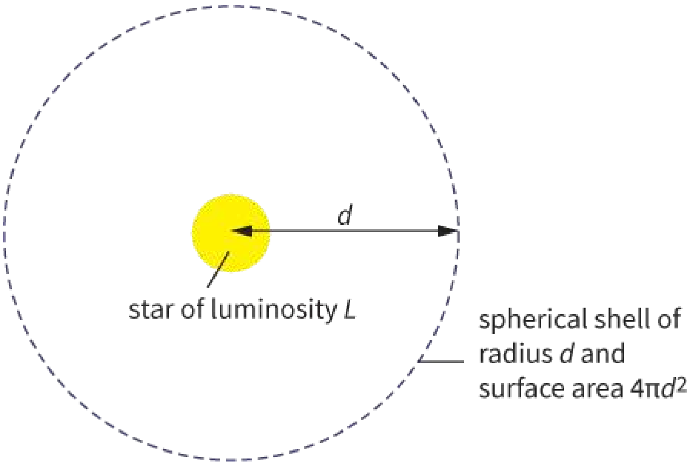

Figure 31.3: The power of the star spreads out uniformly through a spherical shell.

Distance of galaxies

Astronomers on the Earth can determine the radiant flux intensity F of a distant star. The equation $F = \frac{L}{4\pi d^2}$ can then be rearranged to determine the distance d of a star of known luminosity L, for example, a standard candle such as a Cepheid variable star in a distant galaxy.

Now look at Worked examples 1 and 2. In Worked example 1, the radiant flux intensity from the Sun at the Earth is calculated. In Worked example 2, the distance of a star in the Andromeda galaxy is calculated from its radiant flux intensity at the Earth.

WORKED EXAMPLES

1 The radius of the Sun is 6.96×10^8 m and its luminosity is 3.83×10^{26} W.

The orbital radius of the Earth is 1.50×10^{11} m.

Calculate the radiant flux intensity at the surface of the Sun and at the position of the Earth.

Step 1 Calculate the radiant flux intensity at the Sun's surface:

$$\begin{aligned} F &= \frac{L}{4\pi d^2} \\ &= \frac{3.83 \times 10^{26}}{4\pi \times (6.96 \times 10^8)^2} \\ &= 6.29 \times 10^7 \, \mathrm{W m^{-2}} \end{aligned}$$

Step 2 Calculate the radiant flux intensity at the Earth's position.

We can do this using the inverse square law relationship between F and d.

The distance *increases* by a factor of:

$$\frac{1.50 \times 10^{11}}{6.96 \times 10^8} = 215.52$$

Therefore, F will decrease by a factor of 215.52^2

So,

$$\begin{aligned} F &= \frac{6.29 \times 10^7}{215.52^2} \\ &= 1.35 \times 10^3 \, \mathrm{W\ m^{-2}} \end{aligned}$$

Note: An alternative would be to just use $F = \frac{1}{4\pi d^2}$, with $L = 3.83 \times 10^{26}$ W and $d = 1.50 \times 10^{11}$ m. Try it, you will get the same answer. You do not need the radius of the Sun to get the right answer.

2 The radiant flux intensity, measured at the Earth, from a Cepheid variable star in Andromeda is 1.4×10^{-16} W m^{-2}. The luminosity of the star is 1.0×10^{30} W.

Calculate the distance of this star.

Step 1 Rearrange the equation for radiant flux intensity.

$$F = \frac{L}{4\pi d^2} \Rightarrow d = \sqrt{\frac{L}{4\pi F}}$$

Step 2 Substitute and calculate the distance of the star.

$$\begin{aligned} d &= \sqrt{\frac{1.0\times10^{30}}{4\pi\times1.4\times10^{-16}}} \\ &= 2.4\times10^{22}\,\text{m} \end{aligned}$$

This distance is equivalent to 2.5 million light-years.

PRACTICAL ACTIVITY 31.1

Inverse square law for radiant flux intensity

We can simulate the inverse square law nature of light spreading from a star using a bright filament lamp and a light-meter. Commercial light-meters are not calibrated to show radiant flux intensity F in W m^{-2}. Light-meters measure a quantity known as illuminance. We can assume that illuminance, often in a unit known as lux, is directly proportional to radiant flux intensity.

Carry out the experiment in a darkened room.

Measure the illuminance at various distances d from the centre of the lamp.

Since radiant flux intensity is inversely proportional to d^2, and directly proportional to illuminance, a graph of illuminance against $\frac{1}{d^2}$ will be a straight line through the origin.

In a laboratory, there will always be some reflection of light from the walls and ceiling. The best place and time for the experiment is outdoors at night!

Questions

Where necessary, take:

$L_\odot = 3.83 \times 10^{26}$ W

1 ly ≈ 9.5×10^{15} m

1 State **two** factors that affect radiant flux intensity from a star.

2 The radiant flux intensity F of light from a lamp at a distance of 10 cm is 0.32 W m^{-2}. Calculate F from the same lamp at a distance of 15 cm. State any assumption(s) you make.

3 Use data from Table 31.1 to determine, to two significant figures:

- **a** the distance of Sirius from the Earth in metres.
- **b** the luminosity (in W) of
  - **i** Canopus
  - **ii** Vega.
- **c** the radiant flux intensity measured at the Earth from:
  - **i** Sirius
  - **ii** Alpha Centauri.

4 This question is about Sirius and Arcturus.

With the help of calculations and data from Table 31.1, show that Sirius is brighter than Arcturus.

5 The radiant flux intensity from a star measured at the Earth is 2.7×10^{-9} W m^{-2}. The luminosity of the star is 1300 $L_\odot$.

Calculate the distance of this star from the Earth in metres.

31.3 Stellar radii

The Sun can be seen as a glowing ball of gas in the sky. If you briefly look at the Sun through a special filter, like a welder's helmet, you can identify it as a yellow disc in space. The Sun is enormous – it only looks small because it is far away from us. We can determine the diameter of the Sun fairly easily (see Practical Activity 31.2). However, when we look at stars in the night sky, they appear as tiny specks of light – there is no disc to be seen (Figure 31.4). The stars are just too far away. Even the closest stars viewed through powerful telescopes appear as specks of light.

How can astronomers determine the size of stars? In this topic, you will see how two simple laws can be used to determine stellar radii.

Figure 31.4: Even the closest stars appear as specks of light – so how do astronomers determine their size?

PRACTICAL ACTIVITY 31.2

The diameter of the Sun

You can estimate the diameter of our closest star, the Sun, using a simple pin-hole camera. You can make this camera using a shoe-box. One end of the box has a sheet of darkened paper (or aluminum foil) with a tiny hole made with a sharp pin. The opposite end of the box has a sheet of tracing paper, which acts as a screen. A circular image of the Sun is formed on the screen when the camera is pointed towards the Sun. See Figure 31.5.

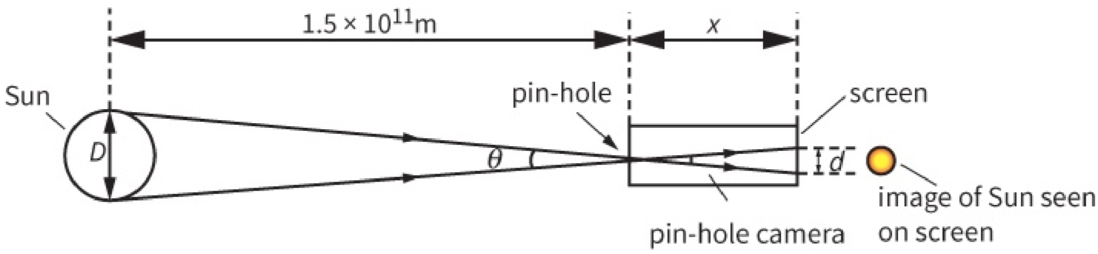

Figure 31.5: You can determine the diameter D of the Sun using a simple pin-hole camera.

Measure the distance x between the pin-hole and the screen. The distance of the Sun from the Earth is 1.5×10^{11} m. The diameter D of the Sun can be determined using simple trigonometry:

$$\tan\theta \approx \frac{d}{x} \approx \frac{D}{1.5\times10^{11}}$$

Therefore:

$$D = \frac{1.5\times10^{11}\times d}{x}$$

Question

6 **a** A student conducted the experiment from Practical Activity 31.2. The results from the experiment are shown below:

x = 300 mm d = 3 mm

Use this data to estimate the diameter of the Sun.

b The actual value for the diameter of the Sun is 1.4×10^9 m.

Determine the percentage difference between your calculated value and the actual value.

Wien's displacement law

The hottest stars are blueish-white in colour. Cooler stars are a deep shade of red. We can see almost the same effect with the filament of a lamp. Increase the temperature of the filament by increasing the current in the filament. At first, the filament will glow dull red when it is cooler, then reddish-orange, and eventually white as it gets hotter.

There is a link between the observed wavelength of light and temperature. Table 31.2 shows the colour of a star in the night sky and the range of its surface temperature.

A hot object, such as a star, can be modelled as a **black body**. A black body is an idealised object that absorbs all incident electromagnetic radiation falling on it. It has a characteristic emission spectrum and intensity that depend only on its thermodynamic temperature. Figure 31.6 shows typical intensity against wavelength graphs for objects at different temperatures.

Colour of star	Surface temperature of star / K
blue	Greater than 33 000
blue to blue-white	10 000 – 30 000
white	7500 – 10 000
yellowish white	6000 – 7500
yellow	5200 – 6000
orange	3700 – 5200
red	Less than 3700

Table 31.2: The observed colour of a star is related to its temperature.

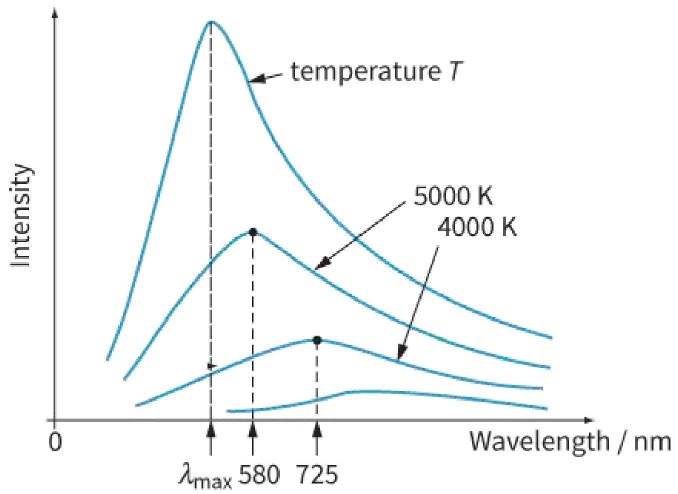

Figure 31.6: The intensity–wavelength graph depends on the temperature of the object. For an object at a thermodynamic temperature T, the intensity against wavelength curve peaks at a wavelength λ_{max}.

The higher the temperature of a body:

- the shorter the wavelength at the peak (maximum) intensity
- the greater the intensity of the electromagnetic radiation at each wavelength.

In 1893, German physicist Wilhelm Wien discovered a relationship between the thermodynamic temperature T of the object and the wavelength λ_{max} at the peak intensity:

$$\lambda_{max} T = \text{constant}$$

The relationship is known as **Wien's displacement law** . The experimental value of the constant is 2.9 × 10^{-3} m K.

The surface temperature of the Sun is 5800 K. This gives a λ_{max} value of about 5.0 × 10^{-7} m or 500 nm. Light of this wavelength appears yellow (which is not surprising for the Sun).

Questions

7 Use the data given in Figure 31.6 to show the validity of Wien's displacement law for 5000 K and 4000 K.

8 For a temperature of 5800 K, the wavelength at peak intensity of electromagnetic radiation is 500 nm. Calculate the surface temperature of a star with wavelength 350 nm at peak intensity.

9 Copy this table.

Star	**Surface temperature T / K**	**λ_{max} / nm**
Sun	5800	500
Polaris	6000	
Canopus	7000	
Gacrus		810

Use Wien's displacement law to complete the table. Write your answers to two significant figures.

The Stefan-Boltzmann law

A quick inspection of Table 31.1 shows that the luminosity of a star does not depend just on the surface temperature of the star. Luminosity also depends on the physical size of the star–its radius. For example, the super red giant star KY Cygni has a surface temperature of 3500 K but its luminosity is 200 000 times that of our Sun. KY Cygni is cooler than the Sun, but its large surface area makes it very luminous.

The luminosity of a star depends on two factors:

- its surface thermodynamic temperature T
- its radius r.

In 1879, Slovenian physicist Josef Stefan developed an expression for the luminosity L of a star. This is the **Stefan-Boltzmann law**:

$$L = 4\pi\sigma r^2 T^4$$

KEY EQUATIONS

Wien's displacement law:

$$\lambda_{max} T = \text{constant}$$
$$\lambda_{max} \propto \frac{1}{T}$$

Stefan-Boltzmann law:

$$L = 4\pi\sigma r^2 T^4$$

where σ is a constant known as the Stefan-Boltzmann constant. The experimental value for σ is 5.67 × 10^{-8} W m^{-2} K^{-4}.

Using Wien's displacement law and the Stefan-Boltzmann law to determine stellar radii

The radius of a star can be calculated from Wien's displacement law and the Stefan-Boltzmann law. The procedure would be as follows:

1 Use Wien's displacement law to determine the temperature T of the star. This would involve determining the wavelength γ_{max} at maximum intensity for the star, and then using a reference star (such as the Sun) to determine T.

2 Use the Stefan-Boltzmann law to determine the radius r of the star. The luminosity L of the star can be determined by measuring the radiant flux intensity F of the star.

The procedure is illustrated in Worked example 3.

WORKED EXAMPLE

3 The surface temperature of the Sun is 5800 K and wavelength of light at peak intensity is 500 nm. The wavelength at peak intensity for Sirius-B (a white dwarf star) is 120 nm. The luminosity of this star is 0.056 times that of the Sun. The luminosity of the Sun is 3.83×10^{26} W.

Calculate the radius of Sirius-B.

Step 1 Use Wien's displacement law to calculate the temperature of Sirius-B.

$$\lambda_{max} T = \text{constant}$$

$$5800 \times 500 = T \times 120$$

$$T = 24\,167\text{ K} \approx 24\,200\text{ K}$$

Step 2 Use the Stefan-Boltzmann law to calculate the radius of Sirius-B.

$$\begin{aligned} L &= 4\pi\sigma r^2 T^4 \\ r &= \sqrt{\frac{L}{4\pi\sigma T^4}} \\ &= \sqrt{\frac{0.056 \times 3.83 \times 10^{26}}{4\pi \times 5.67 \times 10^{-8} \times 24167^4}} \\ &\approx 9.4 \times 10^6\,\text{m} \end{aligned}$$

Sirius-B is roughly the size of our Earth! It is a very hot star, but not very luminous because of its small size.

Question

10 The luminosity of the star Aldebaran is 520 times that of the Sun. The wavelength of light at peak intensity for Aldebaran is 740 nm and the wavelength of light at peak intensity for the Sun is 500 nm.

a Explain whether Aldebaran is cooler or hotter than the Sun.

b Calculate the ratio:

radius of Aldebaran / radius of the Sun.

31.4 The expanding Universe

The **Big Bang theory** is a model of the evolution of the Universe from an extremely hot and dense state some 13.8 million years ago – the event was called the Big Bang.

The Big Bang was also responsible for the birth of the fabric of space (and time) – this fabric has been expanding ever since then. At the early stages after the Big Bang, fundamental particles (such as quarks) and forces (such as gravitation) came into existence. Subsequent expansion led to cooling and formation of atoms, stars and galaxies. The one question that cosmologists cannot answer (yet) is *why* the Big Bang happened in the first place. There are lots of thoughts and theories, but nothing that can be tested.

In this topic, we will explore evidence for the Big Bang by using the ideas of physics developed in the earlier chapters of this book–notably spectra (Chapter 12) and Doppler effect (Chapter 12).

Hubble's law

Astronomers can see the light from distant galaxies using powerful telescopes. The telescopes can look at the light through a diffraction grating. Analysis of the spectrum of the light from distant galaxies shows that they are all moving away from us. The more distant a galaxy, the faster it moves. How do we know from the spectrum that galaxies are moving away from us (receding)?

We examined the Doppler effect of sound in Chapter 12. The observed wavelength of sound was longer for a receding source and shorter for an approaching source. The same happens with electromagnetic waves. The observed wavelengths of **all** spectral lines from distant galaxies are longer than the ones observed in the laboratory. This is known as **redshift**. Figure 31.7 shows the red-shifting of the absorption spectral lines from a cluster of galaxies some 1 billion light-years away.

The redshift of spectral lines from distant galaxies must imply that all galaxies are receding from us. This is what American astronomer Vesto Slipher discovered in 1917. Another American astronomer, Edwin Hubble, combined his own observations with Slipher's discovery to create **Hubble's law**.

Hubble's law states that the recession speed v of a **galaxy** is directly proportional to its distance d from us.

Therefore,

$$v \propto d$$

or

$$v = H_0 d$$

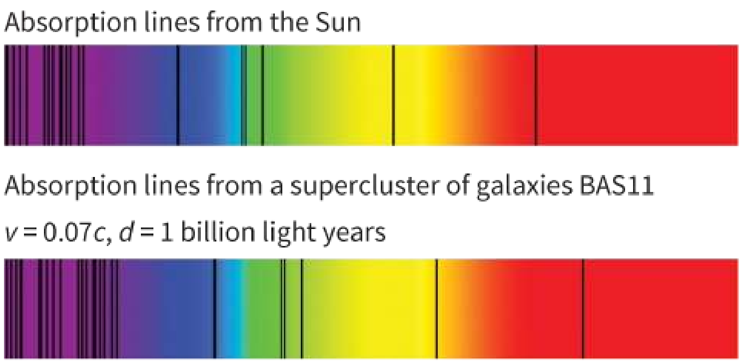

Figure 31.7: The absorption lines in the spectrum of the galaxies are all shifted to longer wavelengths – redshifted. The top spectrum is the spectrum from a 'stationary source', the Sun.

where v is the recession speed, d is the distance of the galaxy and H_0 is the Hubble constant.

The SI unit for H_0 is second^{-1}, or s^{-1}.

The experimental value for H_0 is about $2.4 \times 10^{-18}\ s^{-1}$. Figure 31.8 shows a recession speed v against distance d graph for galaxies. The straight-line of best fit passes through the origin, and the gradient of the line is equal to H_0.

KEY EQUATION

Hubble's law:

$$v = H_0 d$$

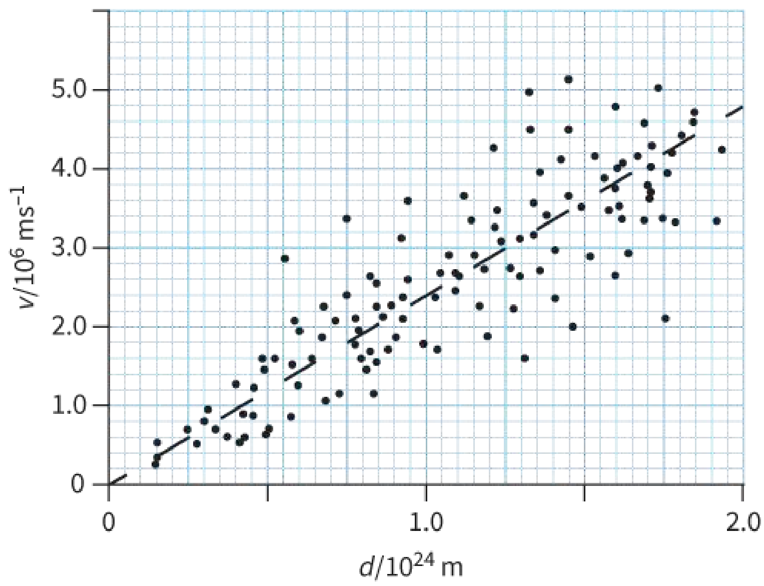

Figure 31.8: Hubble's law shows that recession speed of galaxy ∝ distance from us. The gradient of the best-fit line is equal to H_0 in s^{-1}. The scatter of the data shows considerable uncertainties in the observation.

Question

11 A galaxy is at a distance of 9.5×10^{24} m from us and is moving away with a speed of 2.1×10^7 m s^{-1}.

a Calculate the Hubble constant based on this data.

b Estimate the speed in km s^{-1} of a galaxy at a distance of 1.9×10^{25} m.

Doppler redshift

It is worth pointing out that the term redshift does not imply spectral lines becoming red; all spectral lines show an increase in wavelength. The fractional increase in the wavelength depends on the recession speed v of the source (galaxy).

For non-relativistic galaxies – those moving with speeds far less than the speed of light in a vacuum c – we can use the relationship:

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

where λ is the wavelength of the electromagnetic waves from the source, $\Delta\lambda$ is the change in the wavelength, f is the frequency of the electromagnetic waves from the source, Δf is the change in frequency, v is the recession speed of the source and c is the speed of light in vacuum.

KEY EQUATION

Doppler redshift:

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

Astronomers and cosmologists often assign a value for the term 'redshift'. For example, a galaxy shows redshift of 7.0 % means that:

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c} \approx 0.070$$

Worked example 4 shows how redshift can be used to determine the speed of a distant galaxy.

WORKED EXAMPLE

4 In the laboratory, an emission spectral line is observed at a wavelength of 656.4 nm. The same spectral line, in the spectrum from a distant galaxy, has wavelength 663.1 nm.

Calculate the speed v of the galaxy.

Step 1 Calculate the change in the wavelength of the spectral line.

The observed wavelength is longer; therefore, the galaxy is receding.

$\Delta\lambda = 663.1 - 656.4 = 6.7$ nm

Step 2 Now calculate the speed v using the Doppler redshift equation.

Hint: *You do not need to convert the nm to m, because the ratio* $\frac{\Delta\lambda}{\lambda}$ *will be the same; just make sure you use the* ***same*** *unit for* $\Delta\lambda$ *and* λ.

$$\begin{aligned}
\frac{\Delta\lambda}{\lambda} &\approx \frac{v}{c} \\
\frac{6.7}{656.4} &\approx \frac{v}{3.0\times10^8} \\
0.0102 &\approx \frac{v}{3.0\times10^8} \\
v &\approx 0.0102 \times 3.0 \times 10^8 \\
v &\approx 3.06 \times 10^6 \mathrm{m\ s}^{-1} \\
v &\approx 3.1 \times 10^6 \mathrm{m\ s}^{-1}
\end{aligned}$$

Questions

12 The fractional change in the wavelength of the observed light from a galaxy is 0.15; its redshift is 15 %.

Calculate its recession speed. State any assumptions made.

13 The Tadpole galaxy has a recession speed of 9400 km s^{-1}.

Calculate the fractional change in the wavelength of the observed spectrum.

Evidence for the Big Bang

All galaxies in the Universe are moving away (receding) from each other, and not from the Earth. An observer in another galaxy will reach the same conclusion. The galaxies have motion because space itself is stretching. This is quite difficult to visualise. The best we can do is to imagine the galaxies as dots on the surface of an ever-expanding balloon (see Figure 31.9).

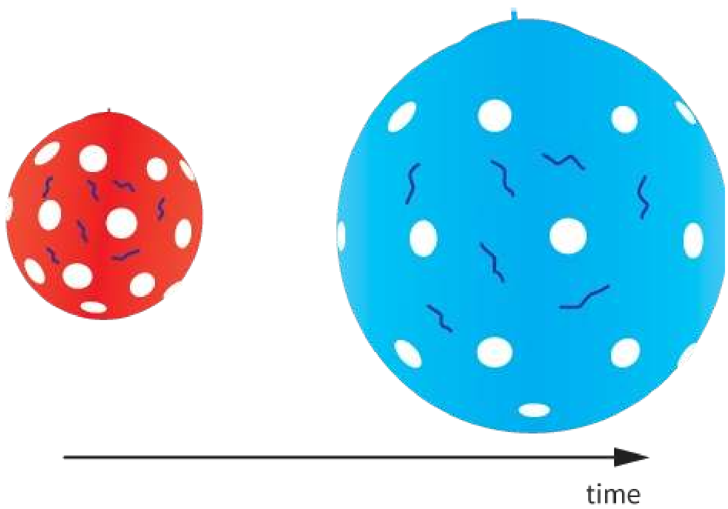

Figure 31.9: The galaxies are modelled as dots on the surface of a balloon. Expansion of the balloon makes all the dots move **away** from each other.

The expanding balloon model can also be used to explain the redshift of light from galaxies. As the Universe expanded, the wavelength of photons was stretched out.

Hubble's law provided the first evidence for the birth, and the subsequent expansion, of the Universe. Distant galaxies appear to be moving faster. However, we must remember that the light has a finite speed, so as we stare deeper into space, we are looking further into the past. The further back in time we go, the faster the galaxies are receding from each other. Rolling back time in this way – like playing a movie in reverse – can only lead to the conclusion that the Universe must have had a beginning … the Big Bang.

How long ago was the Big Bang? We can estimate this from the Hubble constant H_0. In Question 11, the speed of the receding galaxy at a distance of 9.5×10^{25} m was 2.1×10^{7} m s^{-1}. If we assume that this speed has remained unchanged, we can estimate the time when our galaxy and this receding galaxy were at the same place (the time of the Big Bang):

$$\begin{aligned} \text{speed} &= \frac{\text{distance}}{\text{time}} \\ 2.1 \times 10^7 &= \frac{9.5 \times 10^{25}}{\text{time}} \\ \text{time} &= \frac{9.5 \times 10^{25}}{2.1 \times 10^7} \\ &= 4.52 \times 10^{18}\,\text{s} \end{aligned}$$

So, the age of the Universe is roughly 4.5×10^{18} s, or 14 billion years.

Support for the Big Bang theory comes from many other experimental evidences. One of these is worth mentioning here – the temperature of the Universe itself. The expansion of the Universe led to cooling; theories predicted the current temperature of the Universe should be about 2.7 K. Data collected and analysed from telescopes onboard satellites have shown that the peak intensity of the electromagnetic radiation coming from all directions of space occurs at a wavelength of about 1 mm (microwaves). The intensity against wavelength graph is similar to the ones shown in Figure 31.6. According to Wien's displacement law, this corresponds to a temperature of about 3 K.

Physics does make you think. Everything around us, including us, was created during the Big Bang; we could, therefore, suggest that we all have the same age!

Question

14 Use the information given in the table in Question 9 about the Sun to show that the current temperature of the Universe matches with microwaves of wavelength 1 mm at peak intensity.

REFLECTION

Without looking at your textbook, list all the laws from this chapter.

Draw a flow diagram to show how the radius of a star can be determined.

Use the internet to find the most distant object in the Universe and its recession speed.

What was the most important thing you learned personally when working through this chapter?

SUMMARY

Luminosity of a star is defined as the **total** radiant power it emits. Luminosity has the unit watts (W).
Standard candles are used to determine the distance of galaxies. A standard candle is an astronomical object (such as a Cepheid variable star) that has known luminosity.
Radiant flux intensity is defined as the radiant power transmitted normally through a surface per unit area. Radiant flux intensity has the units W m^{-2}. The radiant flux intensity F at a distance d from the centre of a star of luminosity L is given by the equation: $$F = \frac{L}{4\pi d^2}$$
Wien's displacement law: $\lambda_{max}T$ = constant where T is the thermodynamic temperature of the object and λ_{max} is the wavelength at the peak intensity.
The Stefan-Boltzmann law: $L = 4\pi\sigma r^2 T^4$ where σ is a constant known as the Stefan-Boltzmann constant, L is the luminosity of the object (star), r is the radius of the object and T is the surface thermodynamic temperature of the object.
Hubble's law: The recession speed of a **galaxy** is directly proportional to its distance from us. The equation for Hubble's law is: $v = H_0 d$ where v is the recession speed, d is the distance of the galaxy and H_0 is the Hubble constant.
Doppler redshift equation: $$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$ where λ is the wavelength of the electromagnetic waves from the source, $\Delta\lambda$ is the change in the wavelength, f is the frequency of the electromagnetic waves from the source, Δf is the change in frequency, v is the recession speed of the source and c is the speed of light in vacuum.
The Big Bang theory is a model of the creation of the Universe, from an extremely hot and dense state, and its subsequent evolution. The redshift of absorption (or emission) spectral lines from distant galaxies provides evidence for the Big Bang.

EXAM-STYLE QUESTIONS

1 Which statement is correct about radiant flux intensity? **[1]**

A It depends on the area of the measuring device.

B It is measured in W m^{-2}.

C It is the same as luminosity.

D It is the total radiant power emitted from a star.

2 A group of astronomers have determined the radiant flux intensity F from a star and its distance d. The percentage uncertainty in F is 1.2 % and the percentage uncertainty in d is 2.5 %.

What is the percentage uncertainty in the calculated value of the luminosity of the star? **[1]**

A 1.3 %

B 3.0 %

C 3.7 %

D 6.2 %

3 A particular emission spectral line is measured in the laboratory to have a frequency of 7.3×10^{14} Hz.

a Calculate the wavelength of this spectral line in the laboratory. **[1]**

b Calculate the observed wavelength of this same spectral line in the spectrum of a galaxy moving away from the Earth at a speed of:

i 11 Mm s^{-1} **[3]**

ii 7.0 % the speed of light. **[3]**

c The spectrum of all distant galaxies is redshifted. State and explain what you can deduce about the Universe. **[2]**

[Total: 9]

4 **a** State Hubble's law. **[1]**

b The recession speed v against distance d graph for some galaxies is shown.

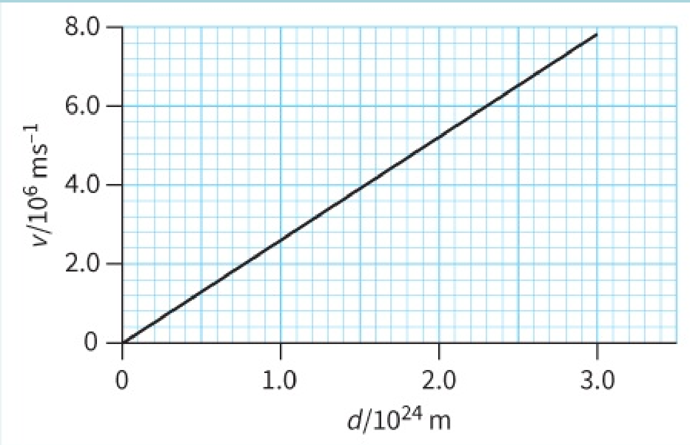

Figure 31.10

Determine the Hubble constant from this graph. Explain your answer. **[3]**

c The Big Bang occurred some 14 billion years ago.

$$1 \text{ year} \approx 3.15 \times 10^7 \text{ s}$$

Estimate the farthest distance we can observe. Explain your answer. **[3]**

[Total: 7]

5 **a** Define the luminosity of a star. **[1]**

b A red giant is a star bigger than our Sun. Explain how the surface of a red giant star can be cooler than the Sun, yet have a luminosity much greater than the Sun. **[2]**

c An astronomer has determined the surface temperature of a white dwarf star to be 7800 K and its radius as 8.5×10^{6} m. Calculate the luminosity of this star. **[3]**

d The surface temperature T of a star depends on the wavelength λ_{max} at the peak intensity of the emitted radiation from the star.

The T against λ_{max} graph for a cluster of stars in our galaxy is shown.

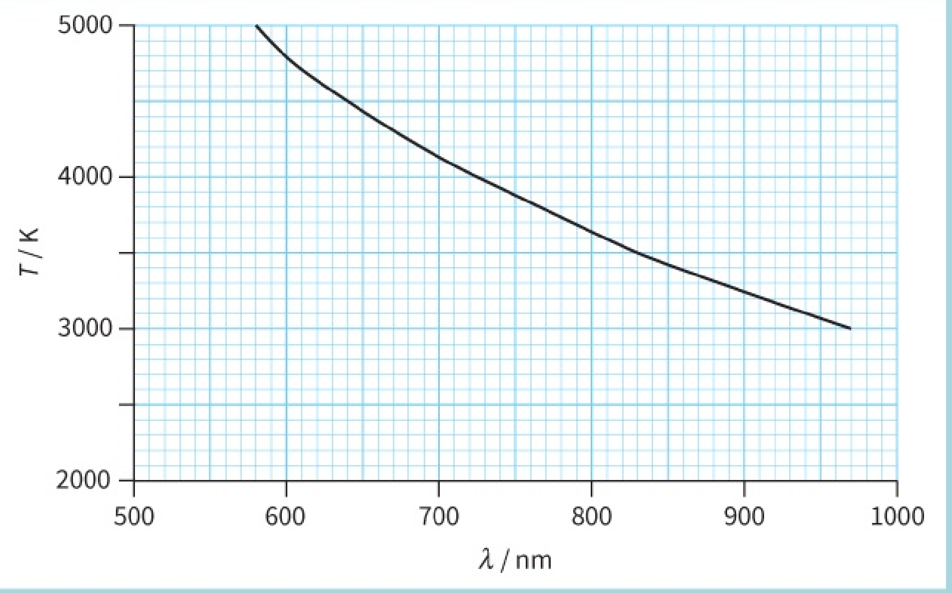

Figure 31.11

i Use the graph to show Wien's displacement law is obeyed. **[2]**

ii Estimate the surface temperature of a star with $\lambda_{max} = 400 \pm 10$ nm. In your answer, include the absolute uncertainty. **[4]**

[Total: 12]

6 Light from a galaxy is passed through a diffraction grating. The diagram shows part of the emission spectrum.

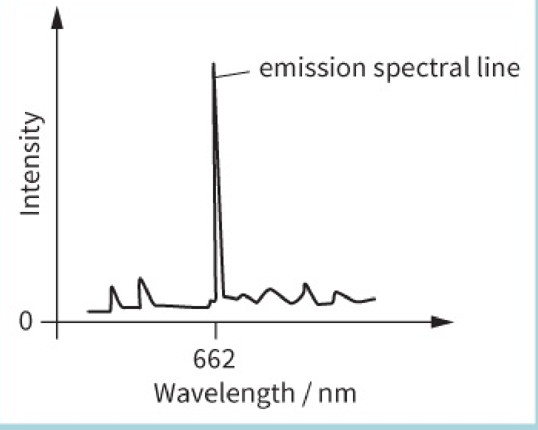

Figure 31.12

The strong emission spectral line has wavelength 662 nm.

a Calculate the energy of a photon of wavelength 662 nm. **[2]**

b Explain how a spectral line is produced by electrons within atoms. **[2]**

c In the laboratory, the same spectral line has wavelength 656 nm.

i Calculate the speed of the galaxy. **[3]**

ii State the direction of travel of the galaxy. **[1]**

d State and explain what the wavelength of the same spectral line would be for a much more distant galaxy. **[2]**

[Total: 10]

7 **a** Define radiant flux intensity. **[1]**

b State the relationship between radiant flux intensity *F* and distance *d* from the centre of a star. **[1]**

c Neptune is the farthest known planet from the Sun in the Solar System. Its distance from the Sun is 30 times greater than the distance of the Earth from the Sun. The radiant flux intensity from the Sun at the Earth is 1400 W m^{-2}.

A space probe is close to Neptune.

Calculate the maximum radiant power received by an instrument of cross-sectional area 1.0 cm^2 on this space probe. **[3]**

[Total: 5]

SELF-EVALUATION CHECKLIST

After studying the chapter, complete a table like this:

I can	See topic...	Needs more work	Almost there	Ready to move on
understand the terms luminosity and radiant flux intensity	31.1, 31.2			
understand the inverse square law nature of radiant flux intensity	31.2			
understand the meaning of the standard candle	31.1			
use Wien's displacement law and the Stefan-Boltzmann law	31.3			
use $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for receding source (stars and galaxies)	31.4			
use Hubble's law	31.4			
understand that redshift of spectral lines from galaxies provides evidence for the Big Bang theory.	31.4			